



University of Asia Pacific

Department of Computer Science and Engineering

Lab Report 2

CMOS NAND and NOR Gates

Assignment: Combined CMOS NAND & NOR Gate Design, Simulation, and Symbol Creation

Course name: VLSI Design Lab

Course Code: CSE 450

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1. Pre-Lab Work

1.1 Objective

The objective of this lab is to study the structure and operation of two-input static CMOS NAND and NOR gates, determine suitable transistor sizing, create the circuits in Cadence Virtuoso, verify their DC and transient responses, and create reusable schematic symbols.

1.2 MOSFET Fundamentals

A MOSFET is a voltage-controlled switching device with four terminals: Gate, Drain, Source, and Body. CMOS logic uses complementary NMOS and PMOS devices.

- NMOS (n-channel): turns ON when V_{GS} exceeds the NMOS threshold and provides a conducting path toward VSS.
- PMOS (p-channel): turns ON when V_{SG} exceeds $|V_{Tp}|$ and provides a conducting path toward VDD.
- In static CMOS, PMOS devices form the pull-up network (PUN) and NMOS devices form the pull-down network (PDN).

1.3 CMOS NAND Gate Structure and Operation

A two-input CMOS NAND gate implements $Y = \neg(A \cdot B)$. The pull-down network contains two NMOS transistors in series, while the pull-up network contains two PMOS transistors in parallel.

- $A = 0, B = 0$: both NMOS OFF and both PMOS ON $\rightarrow Y = 1$.
- $A = 0, B = 1$: the series NMOS path is interrupted $\rightarrow Y = 1$.
- $A = 1, B = 0$: the series NMOS path is interrupted $\rightarrow Y = 1$.
- $A = 1, B = 1$: both NMOS ON and both PMOS OFF $\rightarrow Y = 0$.

1.4 CMOS NOR Gate Structure and Operation

A two-input CMOS NOR gate implements $Y = (A+B)'$. The pull-down network contains two NMOS transistors in parallel, while the pull-up network contains two PMOS transistors in series.

- $A = 0, B = 0$: both NMOS OFF and both PMOS ON $\rightarrow Y = 1$.
- $A = 1, B = 0$: NMOS pull-down conducts and the PMOS series path is broken $\rightarrow Y = 0$.
- $A = 0, B = 1$: NMOS pull-down conducts and the PMOS series path is broken $\rightarrow Y = 0$.
- $A = 1, B = 1$: both NMOS conduct and both PMOS are OFF $\rightarrow Y = 0$.

1.5 Design and Sizing Calculations

For a first-order CMOS sizing approach, R_{ON} is approximately proportional to L/W , so increasing transistor width reduces its effective ON resistance.

For the NAND gate, the two NMOS devices are in series. A common first-order approach is therefore $W_{NMOS,NAND} \approx 2 \times W_{NMOS,ref}$. The PMOS devices are parallel, so their widths can be selected according to the technology model and desired switching characteristics.

For the NOR gate, the PMOS devices are in series. The supplied NOR report uses $W = 240$ nm and $L = 45$ nm for each PMOS, while the NMOS devices use $W = 120$ nm and $L = 45$ nm. Using $kn' \approx 300 \mu A/V^2$ and $kp' \approx 130 \mu A/V^2$, the report gives approximately $kn \approx 800 \mu A/V^2$ and $kp \approx 693 \mu A/V^2$. Two PMOS devices in series give an effective $kp_{eff} \approx 347 \mu A/V^2$ for the worst-case single-input switching event.

Gate	Device	Model	W	L	m	Topology
NAND	NMOS 1/2	Nmos1v	As shown in supplied NAND schematic	As shown	As shown	Series PDN
NAND	PMOS 1/2	Pmos1v	As shown in supplied NAND schematic	As shown	As shown	Parallel PUN
NOR	NM0/NM1	g45n1svt	120 nm	45 nm	1	Parallel PDN
NOR	PM0/PM1	g45p1svt	240 nm	45 nm	1	Series PUN

1.6 Expected Truth Tables (Pre-Lab Prediction)

NAND: $Y = \neg(A \cdot B)$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

NOR: $Y = (A+B)'$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

1.7 Pre-Lab Checklist

- Reviewed NMOS and PMOS switching behavior and CMOS pull-up/pull-down networks.
- Derived NAND and NOR Boolean expressions and truth tables.
- Identified series/parallel transistor topology for each gate.
- Considered sizing for NAND's series NMOS stack and NOR's series PMOS stack.
- Planned the Cadence schematic connectivity and supply connections.
- Predicted the DC logic and transient switching behavior before simulation.

2. Truth Table and Circuit Description

2.1 NAND Truth Table

A (Input)	B (Input)	$Y = \neg(A \cdot B)$ (Output)
0 (LOW)	0 (LOW)	1 (HIGH)
0 (LOW)	1 (HIGH)	1 (HIGH)
1 (HIGH)	0 (LOW)	1 (HIGH)
1 (HIGH)	1 (HIGH)	0 (LOW)

2.2 NAND Circuit Description

The implemented NAND circuit is a static CMOS two-input NAND gate consisting of two PMOS transistors in parallel and two NMOS transistors in series. The PMOS network is connected between VDD and Y, while the NMOS network is connected between Y and VSS/Gnd.

- Both PMOS sources/bodies are connected to VDD and their drains join the output node.

- The NMOS devices form a series path from Y to Gnd.
- A and B control the corresponding PMOS/NMOS gates.
- Only A = B = 1 creates a complete pull-down path, giving Y = 0.

2.3 NOR Truth Table

A (Input)	B (Input)	Y = (A+B)' (Output)
0 (LOW)	0 (LOW)	1 (HIGH)
0 (LOW)	1 (HIGH)	0 (LOW)
1 (HIGH)	0 (LOW)	0 (LOW)
1 (HIGH)	1 (HIGH)	0 (LOW)

2.4 NOR Circuit Description

The implemented NOR circuit is a static CMOS two-input NOR gate consisting of two NMOS transistors in parallel and two PMOS transistors in series. The NMOS network is between Y and Gnd, while the PMOS network is between VDD and Y through the intermediate node X.

- NM0/NM1 sources and bodies are tied to Gnd and drains are tied to Y.
- PM0 source/bulk is tied to VDD; PM0 drain connects to X; PM1 source connects to X; PM1 drain connects to Y; both PMOS bulks are tied to VDD.
- A drives NM0/PM0 and B drives NM1/PM1.
- Y is HIGH only when both inputs are LOW.

3. Circuit Schematics and Simulation Results

3.1 NAND Schematic Capture

The NAND gate was captured in Cadence Virtuoso Schematic Editor using the Nmos1v and Pmos1v devices and the series-NMOS/parallel-PMOS topology specified in the supplied NAND report.

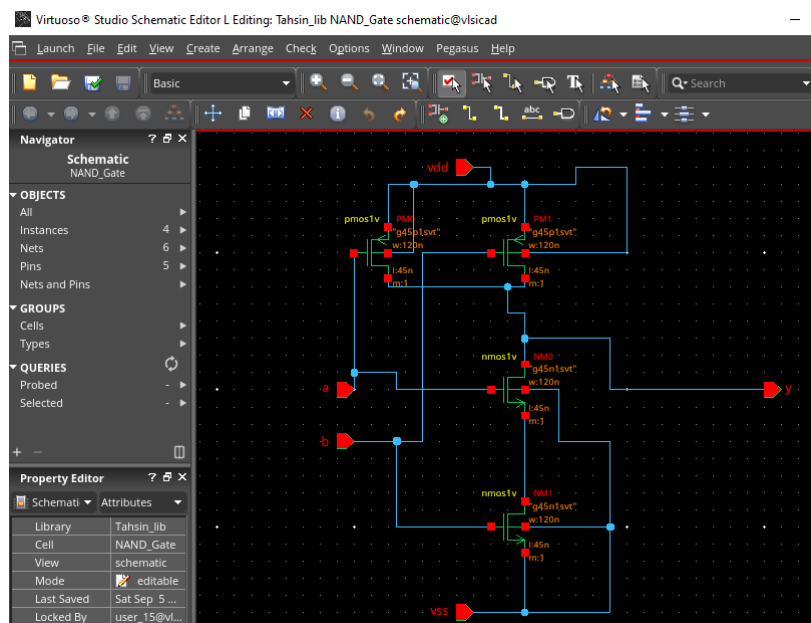


Figure 1. CMOS NAND gate schematic captured in Cadence Virtuoso

3.2 NAND DC / Logic Verification

The DC/logic simulation verifies the output for the four possible input combinations. The expected output is HIGH for 00, 01, and 10, and LOW for 11.



Figure 2. DC/logic verification of the CMOS NAND gate.

3.3 NAND Transient Response

The transient analysis uses time-varying inputs A and B. The output remains HIGH except when both inputs are simultaneously HIGH. Finite rise/fall times may appear because practical MOSFETs have non-zero on-resistance and the output node has capacitance.

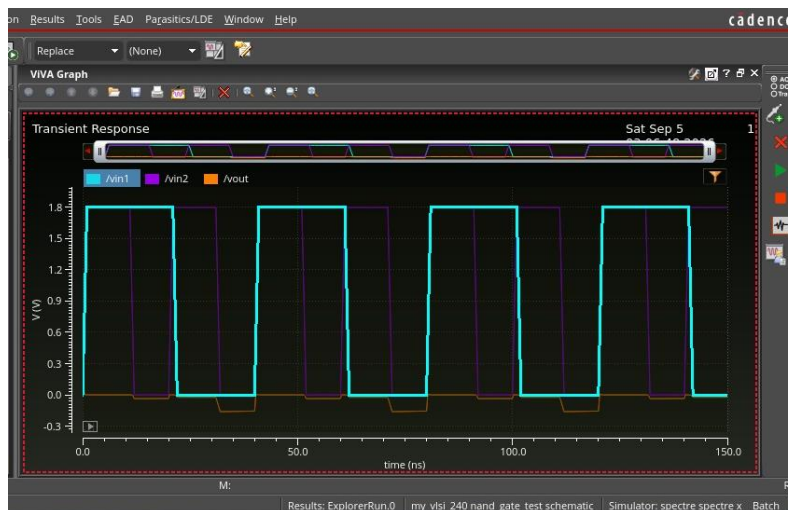


Figure 3. Transient response of inputs A, B and output Y.

3.4 NAND Device Parameter Summary

Parameter	NMOS 1	NMOS 2	PMOS 1	PMOS 2
Model	Nmos1v	Nmos1v	Pmos1v	Pmos1v
Width (W)	As shown in supplied schematic	As shown in supplied schematic	As shown in supplied schematic	As shown in supplied schematic
Length (L)	As shown	As shown	As shown	As shown
Multiplier (m)	As shown	As shown	As shown	As shown
Network	PDN	PDN	PUN	PUN
Connection	Series	Series	Parallel	Parallel

3.5 NOR Schematic Capture

The NOR gate was captured in Cadence Virtuoso Schematic Editor. The supplied NOR report uses NM0/NM1 with model g45n1svt and PM0/PM1 with model g45p1svt.

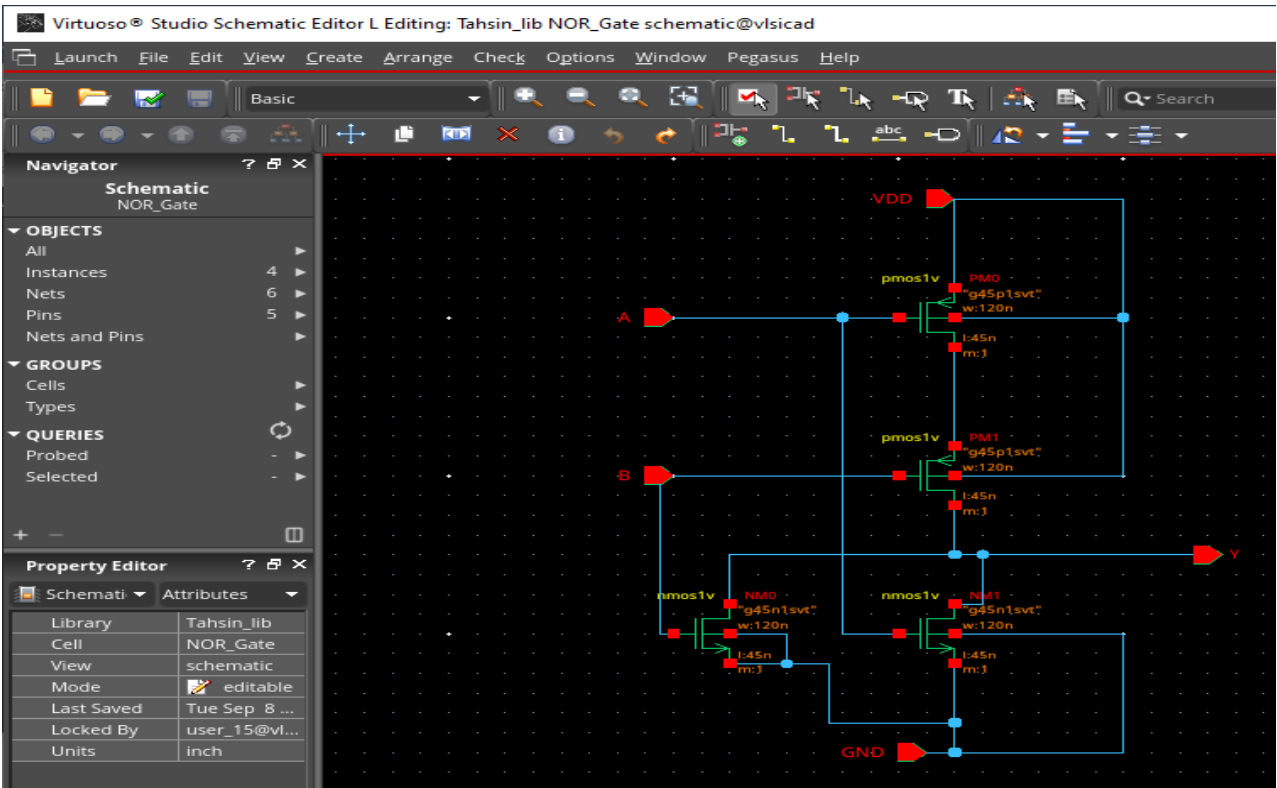


Figure 4. CMOS NOR gate schematic captured in Cadence Virtuoso.

3.6 NOR DC / Logic Verification

The NOR logic verification checks the four possible input combinations. The expected output is HIGH only for 00 and LOW for 01, 10, and 11.

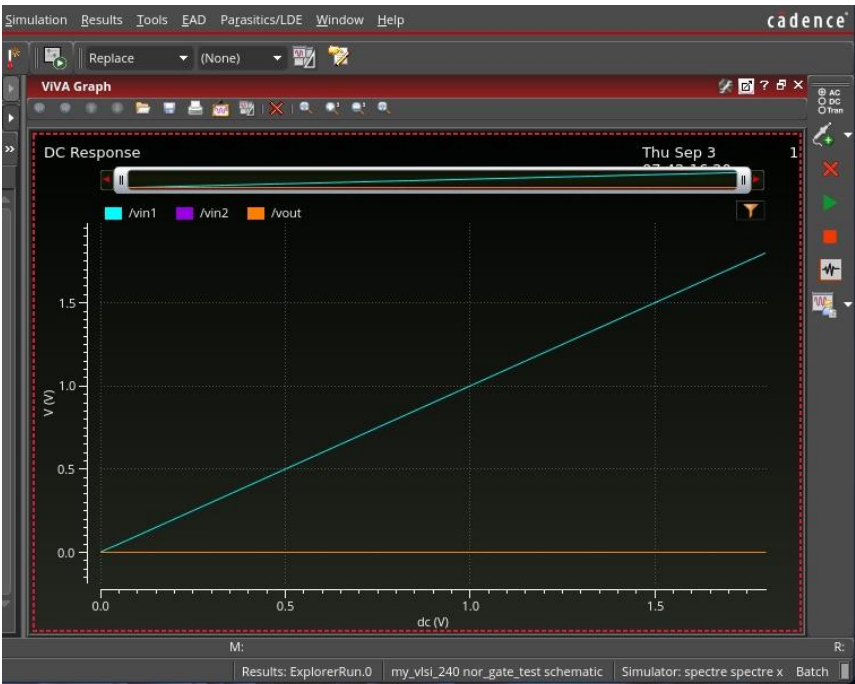


Figure 5. DC/logic verification of the CMOS NOR gate.

3.7 NOR Transient Response

The transient response demonstrates the expected NOR switching behavior, with Y HIGH only while both inputs are LOW.

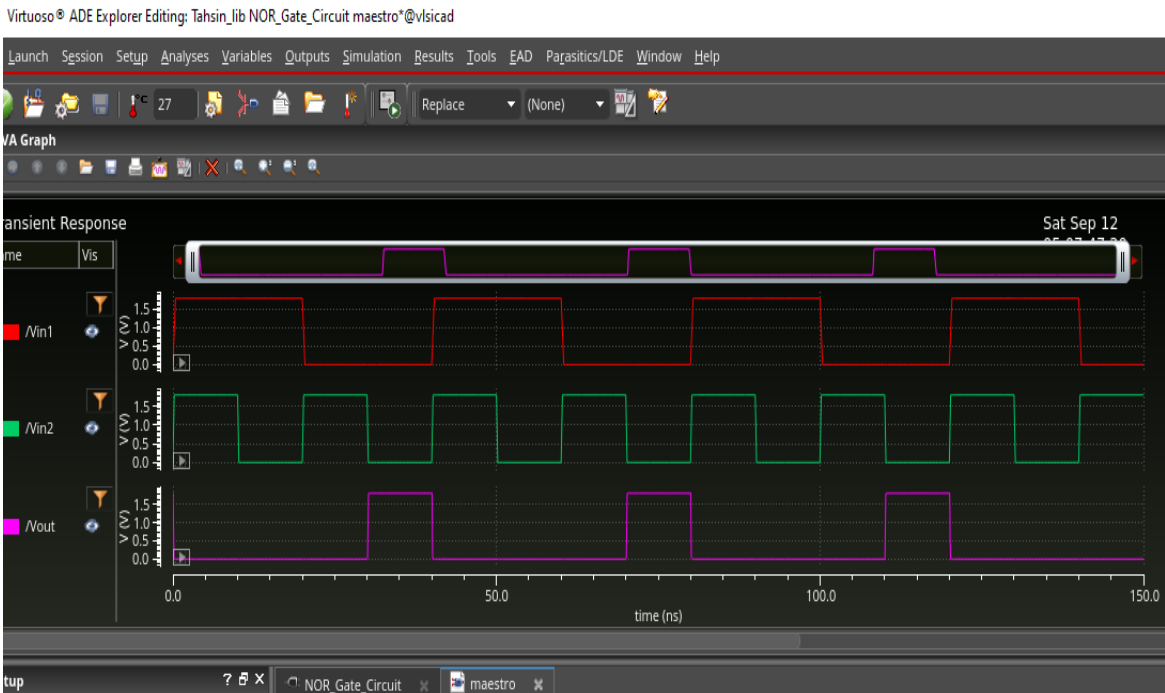


Figure 6. Transient response of inputs A, B and output Y.

3.8 NOR Device Parameter Summary

Parameter	NMOS (NM0/NM1)	PMOS (PM0/PM1)	Remarks
Device model	g45n1svt	g45p1svt	45 nm standard-Vt library
Width W	120 nm	240 nm	PMOS widened for series stacking
Length L	45 nm	45 nm	Minimum channel length
Multiplier m	1	1	Single-finger device
Aspect ratio W/L	≈ 2.67	≈ 5.33	PMOS width = 2× NMOS width
Topology	Parallel PDN	Series PUN	Standard CMOS NOR

3.9 Key Simulation Results

Gate	Logic condition	Expected / Observed
NAND	00	Y = HIGH
NAND	01	Y = HIGH
NAND	10	Y = HIGH
NAND	11	Y = LOW
NOR	00	Y = HIGH
NOR	01	Y = LOW
NOR	10	Y = LOW
NOR	11	Y = LOW

4. Symbol Creation

4.1 NAND Gate Symbol Creation

After transistor-level verification, a schematic symbol was created for the NAND cell so the verified circuit could be reused as a single component in higher-level designs.

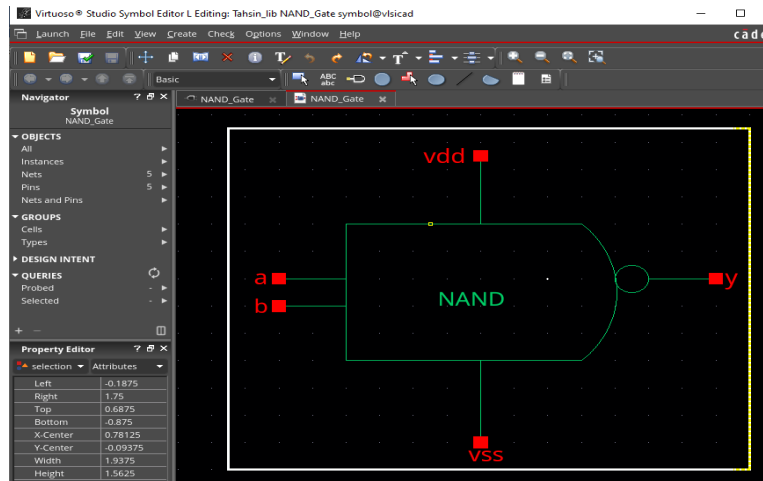


Figure 7. Schematic symbol created for the CMOS NAND cell.

Pin	Direction / Type	Description
A	Input / Signal	First logic input
B	Input / Signal	Second logic input
Y	Output / Signal	NAND output, $Y = \neg(A \cdot B)$
VDD	Input / Power	Positive supply rail connected to PMOS network
VSS	Input / Ground	Ground/return rail connected to NMOS network

4.2 NOR Gate Symbol Creation

After transistor-level verification, a reusable schematic symbol was created for the NOR cell.

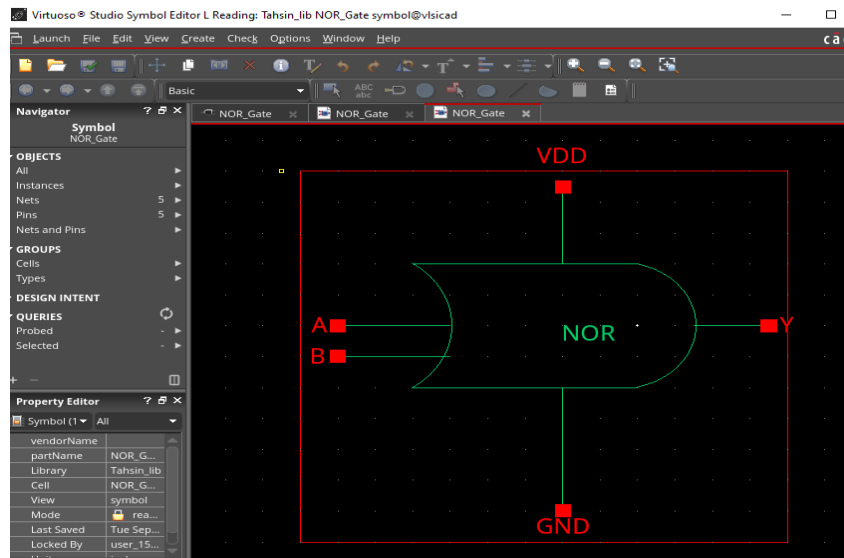


Figure 8. Schematic symbol created for the CMOS NOR cell.

Pin	Direction / Type	Description
A	Input / Signal	First logic input
B	Input / Signal	Second logic input
Y	Output / Signal	NOR output, $Y = (A+B)'$
VDD	Input / Power	Positive supply rail connected to PMOS network
VSS	Input / Ground	Ground/return rail connected to NMOS network

5. Results and Discussion

5.1 NAND Results

- The simulated NAND truth table follows the expected behavior: HIGH for 00, 01, and 10, and LOW for 11.
- The series NMOS network provides the pull-down path only when both inputs are HIGH.
- The parallel PMOS network provides a pull-up path whenever at least one input is LOW.
- Transient switching includes practical propagation delay and finite rise/fall times due to device and load effects.

5.2 NOR Results

- The simulated NOR truth table follows the expected behavior: HIGH only for 00.
- The parallel NMOS network pulls the output LOW whenever either input is HIGH.
- The series PMOS network pulls the output HIGH only when both inputs are LOW.
- The series PMOS stack is weaker than a single PMOS of the same dimensions, which motivates the larger PMOS width used in the supplied NOR design.

5.3 Comparison of Expected and Simulated Logic

Gate	A	B	Expected Y	Simulated Y	Status
NAND	0	0	1	1	Pass
NAND	0	1	1	1	Pass
NAND	1	0	1	1	Pass
NAND	1	1	0	0	Pass
NOR	0	0	1	1	Pass
NOR	0	1	0	0	Pass
NOR	1	0	0	0	Pass
NOR	1	1	0	0	Pass

6. Conclusion

This experiment completed the transistor-level design and verification of two-input static CMOS NAND and NOR gates. The work included pre-lab theory, Boolean-function derivation, truth-table prediction, CMOS pull-up and pull-down network design, transistor sizing, Cadence Virtuoso schematic capture, DC/logic and transient simulation, and reusable symbol creation.

The NAND gate used a parallel PMOS pull-up network and a series NMOS pull-down network, while the NOR gate used a series PMOS pull-up network and a parallel NMOS pull-down network. The simulation results matched the expected truth tables for both gates, confirming correct logical functionality. The experiment also provided practical understanding of CMOS network topology, device sizing, switching behavior, and hierarchical cell reuse in Cadence Virtuoso.

Appendix: Source Figure Placement

The figures in this combined report were extracted directly from the supplied NAND and NOR laboratory documents and inserted into the corresponding schematic, simulation, and symbol sections. No schematic or simulation screenshot was replaced by a newly drawn approximation.